Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/06/2013
PI Name:	Mednieks, Maija Ph.D.		
Project Title:	Protein Expression in Salivary Glands: Effects of Extended Space Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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City:	Farmington	**State:**	CT
Zip Code:	06030-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	06/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; Bion M1 NOTE: Extended to 6/30/2015 (from 12/31/2014), per NSSC information (Ed., 12/16/14) NOTE: To be extended to 12/31/2014, per A. Chu/ARC (Ed., 5/30/14) NOTE: Extended to 7/31/14 per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hand, Arthur D.D.S. (University of Connecticut Health Center)		
Grant/Contract No.:	NNX09AP13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The study was designed to determine effects of extended space travel on oral tissues using the salivary glands of mice. Rodent salivary glands are structurally and functionally similar to those of humans therefore useful for research on changes associated with travel in space. Salivary glands and oral fluid play an important role in maintaining the integrity of the mouth and teeth. Proteins present in secretion granules of salivary glands are released into saliva and can be used to monitor effects of environmental stress. Salivary proteins can serve as markers for physiologic processes such as digestive and immune systems and hormonal regulation. The proteins in saliva are reliable indicators of the environmental effects of travel in space. The physiologic processes for which they are markers are either slowed, accelerated or unaffected. Several of the proteins have previously been shown to respond to a variety of environmental stimuli, including space flight. Our previous studies on short duration (~ 1 week) space flight missions showed that specific secretory proteins were modified by exposure to both zero (ØG) and hyper G conditions. Proteins associated with hormonal signaling and gene expression are down-regulated. An enzyme that is associated with digestion seems unaffected while some cellular proteins appear increased. Employing saliva as a diagnostic test medium could have clinical applications and be useful in measuring human responses to space flight. The present results agree with previous findings that proteins in salivary glands are specifically altered during space flight and can therefore be used as markers for environmental effects on different physiological and behavioral responses
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Background and Objectives: It is important to study the effects of travel in space and to measure reactions to the stress of weightlessness in zero gravity (ØG) conditions. Stress changes hormone action that is reflected in glands that produce saliva as well as in different body parts/organs. The goal is to devise a clinical test for stress disorders in space and on Earth. Saliva contains substances (e.g. proteins) that change during travel in space. We can determine which salivary proteins have been changed by using biochemical tests and histologic analyses to study tissues and cells (salivary glands) that produce saliva in mice flown on the space shuttle Discovery STS-131, 133, and 135 missions. This information will be used to design a test kit to measure responses in astronauts so no blood will have to be drawn. Collecting saliva instead of blood and urine, will be easier during space flight and will make it possible to study hormone changes experienced by Astronauts and Cosmonauts at ØG. Travel in space is an ideal model system for studying a specific environmental impact on physiologic function that is due to gravitational change and cannot be duplicated on Earth. Understanding the effect of ØG will define changes due to the environment as the stressor. This information can then be adapted to devising specific tests of different conditions using saliva for general clinical applications in medicine in space or on Earth. Research Description: The Biospecimen Sharing Program has provided an unparalleled opportunity to study tissue and secretory protein signaling in tissues affected by environmental stimuli whereby cellular and molecular changes associated with travel in space can be identified in experimental animals and can be monitored by measuring the expression of specific proteins. Previous studies on the effects of space flight have shown that regulation by catecholamine hormones is markedly altered during space flight. The expression of specific proteins as markers can potentially be used for measuring responses to environmental stress since changes in the expression of secretory proteins are reflected in saliva.
Task Progress:	The effects of microgravity conditions are highly specific. Different salivary compounds, in this case proteins, represent a specific function. So, changes during space flight in an enzyme in saliva that digests complex carbohydrates can be a signal that nutritional function is changed. Another protein that is secreted from the glands into saliva that functions in immune responses appears unchanged during space travel and indicates that immune reactions may not be affected at ØG. These findings can be used to establish saliva as a diagnostic tool to assess physiological responses of astronauts as well as for clinical applications on Earth. Recent data from the last flight of Discovery show that differences in protein expression in flight samples begin returning to normal by day 5 after landing. Data were collected for the long-term purpose of using salivary protein changes to study physiological changes in astronauts, and ultimately in Earth-based studies. The relationship between protein expression in oral tissues and exposure to a low-gravity environment is not known; however, previous studies have shown that environmental changes can alter gene expression. Most recent flight samples were collected from the Russian BION M-1 spacecraft that had a flight duration of 30 days. Experiments will be conducted to determine if the changes seen in shorter flights intensify or the system adapts and the changes in long duration spaceflight diminish.
Bibliography Type:	Description: (Last Updated: 02/04/2021)
Abstracts for Journals and Proceedings	Mednieks MI, Tsesis M, Larson N, Hand AR. "STS-131: Protein Expression in Heart and Salivary Glands: Effects of Extended Space Flight." IADR/AADR/CADR 89th General Session and Exhibition, San Diego, CA, March 16-19, 2011. Session Chair M. Mednieks. J Dent Res. 2011;90(Spec Iss B):abstract number 2078. http://www.dentalresearch.org , Mar-2011
Abstracts for Journals and Proceedings	Larson NA, Mednieks MI, Hand AR. "Effect of spaceflight on protein expression in mouse submandibular glands." American Academy of Pediatric Dentistry 2011 Meeting, New York, NY, May 2011. Poster presentation. American Academy of Pediatric Dentistry 2011 Meeting, New York, NY, May 2011. , May-2011
Abstracts for Journals and Proceedings	Khatri A, Tsesis M, Manz D, Perrier J, Mednieks MI, Hand AR. "Protein expression in mouse parotid gland: Effects of spaceflight." Annual Biomedical Research Conference for Minority Students, St. Louis, MO, November 2011. Platform presentation. Annual Biomedical Research Conference for Minority Students, St. Louis, MO, November 2011. , Nov-2011
Abstracts for Journals and Proceedings	Khan I, Mednieks M, Hand AR. "The effect of space travel on mouse salivary gland protein expression. University " University of Connecticut Health Center Student Research Day, Farmington, CT, February 2012. Platform presentation student research competition. University of Connecticut Health Center Student Research Day, Farmington, CT, February 2012. , Feb-2012

Abstracts for Journals and Proceedings	Rubenstein R, Burleson J, Mednieks M, Hand A. "The effect of microgravity on secretory protein expression in mouse parotid glands." University of Connecticut Health Center Student Research Day, Farmington, CT, April 2012. Student Research competition. University of Connecticut Health Center Student Research Day, Farmington, CT, April 2012. , Apr-2012
Abstracts for Journals and Proceedings	Mednieks M, Rubenstein R, Burleson J, Hand A. "The Effect of Microgravity on Secretory Protein Expression." 19th IAA Humans in Space Symposium, Cologne, Germany, July 7-13, 2013. Poster presentation in Section on Molecular Medcine. 19th IAA Humans in Space Symposium, Cologne, Germany, July 7-13, 2013. Abstract #428. , Jul-2013
Awards	Mednieks M, Hand A. "Connecticut Space Grant Consortium Annual Award, April 2012." Apr-2012
NASA Technical Documents	Mednieks MI, Hand AR. "STS -133 30-Day Post Flight Operational Report." Moffett Field, CA : NASA ARC Biosciences Division, 2010. , May-2010